Problem set 2 solution

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1 Problems Set 2

Problem 1.1 (10pt)**.** Let $\mathbb{F}$ be an ordered field with $1 \neq 0$. Show that $1 > 0$. *Hint: show* $(-1)(-1) = 1$ *first.*

Problem 1.2 (10pt)**.** Recall that two rational numbers $\frac{n_1}{m_1}$ and $\frac{n_2}{m_2}$ are the same, denoted by $\frac{n_1}{m_1} = \frac{n_2}{m_2}$, if $n_1 m_2 = n_2 m_1$. Show that the addition

$$\frac{n}{m} + \frac{p}{q} := \frac{nq + mp}{mq}$$

is well-defined. That is, if $\frac{n_1}{m_1} = \frac{n_2}{m_2}$ and $\frac{p_1}{q_1} = \frac{p_2}{q_2}$, then

$$\frac{n_1}{m_1} + \frac{p_1}{q_1} = \frac{n_2}{m_2} + \frac{p_2}{q_2}.$$

Problem 1.3 (10pt)**.** Find $\sup E$ and $\inf E$ for the following sets E:

1. $E = \{n \in \mathbb{Z} \mid n < \sqrt{12}\}$
2. $E = \{r \in \mathbb{Q} \mid r < \sqrt{12}\}$
3. $E = \{x \in \mathbb{R} \mid x^2 - x - 1 < 0\}$
4. $E = \left\{ \frac{n^2+n}{n^2+1} \,\middle|\, n \in \mathbb{N} \right\}$

No proof is needed for the answer.

Problem 1.4 (10pt)**.** Let $\mathbb{M}$ be the set of polynomials with integer coefficients,

$$\mathbb{M} := \{f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \mid a_i \in \mathbb{Z}\}.$$

Define the relation $0 \prec f$ if $0 < f(x)$ for x large enough. To be more precise, we say $0 \prec f$ if there exists $M > 0$ such that $f(x) > 0$ for all $x > M$. Then define $f \prec g$ if $0 \prec g - f$. Show that $(\mathbb{M}, \prec)$ is an ordered set.

You can use the following fact directly (without proving it): if $f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$ with $0 < a_n$, then $0 < f(x)$ for x large enough.

Problem 1.5 (10pt)**.** Let $(\mathbb{M}, \prec)$ be the ordered set defined in Problem 4. Show that $(\mathbb{M}, \prec)$ **doesn't** satisfy the Archimedean property.

Problem 1.6 (10pt)**.** The greatest lower bound of a set E is defined to be the number β which has the following properties:

- For all $x \in E$, $\beta \le x$.
- Suppose $\alpha \le x$ for all $x \in E$. Then $\beta \ge \alpha$.

Show that for any non-empty set $E \subset \mathbb{R}$ which is bounded from below, E has the greatest lower bound.

Proof. Let $F = \{-x \mid x \in E\} \subset \mathbb{R}$, then F is bounded from above, let $a = \sup F \therefore -x \ge a$ so that $x \ge -a$ for all $x \in E$, then $-a$ is the lower bound of E.

Assume that there are b such that b is the lower bound of E. $\therefore x \ge b \Longrightarrow -x \le -b$ then $-b$ is the upper bound of F. Since $a = \sup F$ then $a \le -b$ then $-a \ge b$. Then $-a$ is the greater upper bound of E. therefore, E has largest lower bound. □

Problem 1.7 (20pt)**.** Show that for all real numbers $x \in \mathbb{R}$, there exists a real number $y \in \mathbb{R}$ such that $y^3 = x$. *Hint: start with the case* $x > 0$.

Proof. WLOG, we let $x > 0$ Let $A = \{a \subset \mathbb{R} \mid 0 < a^3 < x\} \subseteq \mathbb{R}$ since A is bounded, there the there's a sup. Let $y = \sup A$, we'll show that $y^3 = x$.

Assume that $y^3 > x$, choose $h > 0$ such that $0 < h < \dfrac{y^3 - x}{3y^2}$. therefore

$$(y-h)^3 = y^3 - 3y^2h + 3yh^2 - h^3 > x$$

so that $y - h$ is upper bound of A but it contradicts that y is smallest.

Assume that $y^3 < x$, we choose $0 < h < 1$ such that $h < \dfrac{x - y^3}{3y^2 + 3y + 1}$ so that $(y+h)^3 < x$ then $y + h \in A$. Therefore, $y > y + h$ which impossible.

Hence $y^3 = x$ □